NATIONAL CHILDREN'S SCIENCE CONGRESS (NCSC) 2026–27

OFFICIAL FORM-A REGISTRATION DRAFT & PROJECT ABSTRACT

Document 04: Administrative Submission Forms & Standalone Scientific Abstract

Notice to Student: *Form-A is the official registration form required by the Department of Science & Technology (DST) and state coordinators. A physical copy of this filled and signed form must be attached at the front of your project file and submitted online to the NCSC portal. Do not invent official registration numbers, signatures, or official seal stamps.*

PART I: OFFICIAL FORM-A REGISTRATION DRAFT (ANNEXURE-I)

1. State:	Bihar	**District:**	[Name of District in Bihar]
2. Block / Taluka:	[Name of Block / Taluka]	**Area Type:**	Rural (Gramin) [X] Urban [ ]
3. Project Title:	**KRISHI-DRISHTI: An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture**		
4. Focal Theme:	Science and Innovation for Sustainability	**Sub-Theme:**	Food, Agriculture and Health
5. Language Used:	English (Report & Presentation)	**Age Category:**	Senior Group (14 to <17 Years)
6. Institution / School:	[Full Official Name of Secondary / Senior Secondary School]		
7. School Address:	[Village / Ward, Post Office, Block, District], Bihar — PIN: [_____]		
8. School Phone & Email:	Phone: [_____] | Email: [_____]		
9. Group Leader (Student 1):	Name: [Full Name of Student Leader] Gender: Male [ ] Female [ ] Other [ ] Date of Birth: ____/____/2011 (Age: 14 Years) Disability Status: No [X] Yes [ ] (If yes, specify code: ____) Class: Class 9 | Contact Phone: [_____]		
10. Co-Worker (Student 2):	Name: [Full Name of Co-Worker Student] Gender: Male [ ] Female [ ] Other [ ] Date of Birth: ____/____/2011 (Age: 14 Years) Disability Status: No [X] Yes [ ] Class: Class 9 | Contact Phone: [_____]		
11. Guide Teacher:	Name: [Full Name of Guide Teacher] Gender: Male [ ] Female [ ] Designation: [TGT Science / Biology / Physics Teacher] Contact Phone: [_____] | Email: [_____]		

Signature of Group Leader	**Signature of Guide Teacher**	**Head of Institution / Principal**	**District Coordinator NCSC**
_____ Date: ____/____/2026	_____ Date: ____/____/2026	_____ (Official Signature & School Seal)	_____ (Verification Stamp & Seal)

PART II: OFFICIAL NCSC PROJECT ABSTRACT

(Strictly formatted for Senior Group / Class 9; 285 words total — complies with the 250–350 word limit)

Project Title:	KRISHI-DRISHTI: An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture
Focal Theme:	Science and Innovation for Sustainability
Sub-Theme:	Food, Agriculture and Health
Research Question:	Can a low-cost, portable handheld microcontroller wand integrated with local microclimate telemetry and an offline edge-AI engine provide reliable preliminary foliar health screening for smallholder farmers without requiring internet connectivity or cloud servers?

In rural Bihar, over 35% of potato and tomato crops are devastated annually by Late Blight (*Phytophthora infestans*) and Early Blight (*Alternaria solani*). Smallholder farmers (< 2 hectares) cannot access expensive laboratory tests (■15,000+) and cannot run commercial cloud AI apps due to zero rural 4G/5G connectivity, causing widespread panic-spraying of toxic fungicides. To solve this, I developed **KRISHI-DRISHTI**, an ultra-low-cost (< ■1,500), 100% offline-first handheld diagnostic screening tool.

The system integrates: (1) a lightweight controller wand (ESP32 DevKit V1) with active-low buttons, an SSD1306 OLED, and a DHT11 canopy sensor; (2) an optical unit (AI-Thinker ESP32-CAM, OV5640 5MP) with an 8–10 cm cylindrical 3D shroud to eliminate macro blur and diffuse harsh shadows; and (3) a pocket edge-AI engine executing on the farmer's smartphone via a local Wi-Fi SoftAP mesh (192.168.4.1) requiring zero internet. The MobileNetV3-Small neural network was quantized via Float16 post-training optimization to 2.03 MB (SHA-256: be71ededc...), achieving 4.94 ms inference. Cryptographic deduplication reconciled 7,710 records into 6,965 unique training hashes, isolating all benchmarks.

On an independent 60-specimen botanical benchmark, the system achieved 88.33% strict six-class accuracy (53/60) and 92.98% accepted diagnostic accuracy (53/57, 3 gated UNCERTAIN), alongside 90.0% pathogen-family correlation (54/60). On 93 unseen outdoor PlantDoc images, raw accuracy was 33.33% (31/93), elevated to 55.36% (31/56) by dual-stage Shannon entropy gating (H > 1.45 nats). Physical testing revealed that torch drop-shadows caused false alarms on healthy leaves and an alien tea leaf reached 87.0% false confidence, proving that the shroud standardizes optical distance rather than providing out-of-distribution biological immunity. KRISHI-DRISHTI provides an honest preliminary screening tool that aids smallholder farmers in early sustainable intervention.

Keywords: *Edge-AI, Offline Screening, Solanaceae Blight, ESP32, MobileNetV3, Shannon Entropy, Microclimate Risk Advisory.*

PART III: NCSC PRE-SUBMISSION CHECKLIST & SUMMARY EVALUATION SHEET

Child scientists should verify every item below before submitting their project dossier to the district coordinator:

Item #	Official NCSC Requirement	Project Compliance Status	Child Scientist Verification
1	Registration Form-A filled and signed by Head of Institution	Drafted in Document 04; awaiting principal stamp	[ ] Verified
2	Written Project Report within 3,500-word Senior limit	Document 01 body text is ~2,913 words (100% compliant)	[ ] Verified
3	Project Logbook / Daily Diary maintained in real-time	Document 02 provides complete structured template	[ ] Verified
4	Exactly Four Chart-Paper Posters (55 cm × 70 cm)	Document 05 contains blueprints for all 4 posters	[ ] Verified
5	Student Age Eligibility (14 to <17 years as on 31 Dec)	Class 9 student (DOB verified via birth certificate)	[ ] Verified
6	Authentic Local Survey / Experimentation	Conducted on potato/tomato plots in rural Bihar	[ ] Verified
7	Scientific Honesty & Real-World Boundaries	Explicitly reports tea leaf OOD & shadow failure modes	[ ] Verified

Quick Reference Summary for Science Congress Judges:

Parameter	Verified Project Value / Specification
Total Hardware Bill of Materials	< ■1,500 total (ESP32 DevKit ■450, ESP32-CAM ■650, OLED ■220, DHT11 ■95, Shroud ■120)
Connectivity Dependency	100% Offline (Local Wi-Fi SoftAP mesh @ 192.168.4.1; zero cloud API calls; zero SIM)
Model Size & Memory Footprint	2.03 MB (2,123,856 bytes, Float16 PTQ, SHA-256: be71ededc...) — cached in RAM
Edge Inference Latency	4.94 ms on PC reference interpreter; ~12–15 ms on budget mobile CPU (Galaxy A03)
Botanical Benchmark Accuracy	88.33% Strict 6-class (53/60); 92.98% Accepted (53/57); 90.0% Pathogen Match (54/60)
PlantDoc Outdoor Accuracy	33.33% Raw (31/93); 55.36% Accepted Diagnostic (31/56); 39.78% Gated (37/93)
Microclimate Integration	Wallin/Hyre risk criteria (RH >= 85%, 18–24.5°C) suppresses unneeded chemical sprays
Aforementioned Failure Modes	Direct torch drop-shadows mimic necrosis; alien tea leaf false accepted at 87% confidence